

ST211: Applied Regression
End-of-Term Project: Individual Project

How Are Gambling Outcomes Associated with Health and Underlying Socio-Economic, Demographic and Lifestyle Factors?

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Introduction

This report has been commissioned by the NHS and NHS Scotland to analyse data from a cohort survey run in England and Scotland. This dataset is well-suited for identifying significant predictors of problem gambling, as it provides comprehensive data on socioeconomic, lifestyle, health, and demographic factors. This report specifically investigates the impact of health due to the convergence of the literature on the effect of mental health and substance abuse on gambling, while little is known about the effect of physical health (Black et al., 2013). One study details the health habits of problem gamblers in the US, and their tendency to avoid exercise and smoke and drink excessively (Pilver et al., 2013), however these findings cannot be translated to British contexts due to distinct healthcare environments. This highlights how these factors do not stand alone; Messerlian et al. (2005) discuss how such research needs to understand socioeconomic factors and cultural influences. Accordingly, this report details a logistic regression analysis on a cleaned sample of 13,106 English and Scottish individuals, identifying the health factors and confounders associated with problem gambling to be:

Age, Sex, NS-SEC Class, Long-Standing Illness, Cigarette Smoking Status, Units of Alcohol Consumed per Week, and Warwick-Edinburgh Mental Wellbeing Scale.

Exploratory Data Analysis (EDA)

The use of boxplots and stacked bar charts provided preliminary insights into any associations between the various predictors and the binary outcome, problem gambling (*PROBGAM*).

Health Factors

The analysis of health factors was split into binary diagnoses (of diabetes, respiratory, digestive, circulatory and heart ailments, and long-standing illnesses), self-assessed and mental health scores, alcohol and cigarette habits, and physical activity. Of these diagnoses, those who have diabetes (*docinfo1*) and respiratory (*compm8*) conditions are less likely to be problem gamblers. Those with digestive conditions are equally likely to be problem gamblers as those who do not, while those who do not suffer from heart and/or circulatory conditions are less likely to be problem gamblers. With regard to mental health, *wemwbs* and *ghq12scr* report low mental health correlates with problem gambling, while mental health diagnoses (*compm3*) shows no impact. While the effect of mental health on problem gambling is ambiguous, we see an association with substance-related predictors: *cigst1*, recorded cigarette smoking status, shows those who occasionally smoked or currently smoke are more likely to partake in problem gambling, and *drating* reports that problem gamblers are likely to have a high alcohol intake. Finally, predictors concerning physical activity and BMI show those who engage in physical activity at work and exercise monthly are likely to engage in problem gambling. BMI does not reveal substantial differences across values.

Supplementary Factors

To investigate potential confounding factors, non-health factors were analysed. This revealed problem gambling was prevalent among young, single, men (*age*, *maritalg*, *Sex*) of Black African/Caribbean or mixed heritage (*ethnicC*), who followed non-Abrahamic religions (*Religsc*). *eqv5* revealed individuals with low equivalised income were more likely to gamble problematically than those earning highly. This is consistent with *hpnsec5* and *OwnRnt08*, where higher class and outright property ownership were associated with lower rates of problem gambling. *Econact_2* showed higher rates of problem

gambling among the unemployed or those in full-time education, supporting *SrcInc15* and *SrcInc7*; those with no income or on financial support are more likely to gamble problematically. *EducEnd* (age finished education) and *HighQual* (highest qualification) show leaving school around GCSE age or holding only GCSE-level qualifications is associated with higher rates of problem gambling.

Analysis

Initially, a regression was run with all health factors which did not show multicollinearity (VIF analysis). Many of these showed to be non-significant, so backwards elimination resulted in a model with only *docinfo1* (diagnosed diabetes), which proved to be better than the null model. Using forwards selection on predictors identified from the EDA (*longill12*, *cigst1*, *ghq12scr*, *drating*, *wemwbs*, and *bmival*) proved better than the null model. However, only *longill12*, *cigst1*, *drating*, and *wemwbs* were significant. After multiple iterations between *docinfo1* and the selected variables, the significant values from the EDA performed best, so they formed a health-focused model.

Adjusting for Demographic, Socioeconomic, and Lifestyle Confounders

Five confounding factors were investigated: age (*age*), sex (*Sex*), income (*eqvinc*), ethnicity (*ethnicC*), and class (*hpnssec5*). Shaffer and Korn (2002) relay research suggesting gambling is prevalent among men. Also, they show younger people have higher rates of gambling problems and while high-income households spend more on gambling activities, it is of less proportion than lower-income households. Finally, Shaffer and Korn state that there are differing rates between ethnic groups. This literature, combined with the EDA, justifies these variables as potential confounders. A logistic regression model was run on these confounders with VIF analysis showing no multicollinearity. Each confounder was individually tested to assess its isolated effect, and *anova()* was run where possible to compare models. Overall, *age*, *Sex*, and *hpnssec5* proved significant, returning a difference of 241.0.

The Final Model

The final model includes the significant predictors from the health model and *age*, *Sex*, and *hpnssec5* as confounders. The inclusion of *age* rendered *longill12* as non-significant, signifying confounding.

Results

Model Summary Statistics

n (Sample Size)	8352
k (Degrees of Freedom)	13
Residual Deviance	6547.9
Null Deviance	6788.9
δ_d (as Null Deviance – Residual Deviance)	241.0

Regression Coefficients and Significance Levels

Variable	coef.est	coef.se	z value	Pr(> z)
(Intercept)	-0.55	0.23	-2.42	0.02
age	-0.01	0.00	-6.74	0.00
as.factor(Sex)2	-0.65	0.07	-9.91	0.00
as.factor(hpnssec5)2	0.42	0.11	0.00	3.92
as.factor(hpnssec5)3	0.08	0.11	0.68	0.50
as.factor(hpnssec5)4	0.30	0.11	2.59	0.01
as.factor(hpnssec5)5	0.33	0.08	4.04	0.00
as.factor(longill12)2	0.00	0.07	0.05	0.96
as.factor(cigst1)2	0.19	0.14	1.36	0.17
as.factor(cigst1)3	0.01	0.08	0.10	0.92
as.factor(cigst1)4	0.29	0.08	3.43	0.00
drating	0.00	0.00	3.05	0.00
wemwbs	-0.01	0.00	-2.93	0.00

Comments About the Data and Analysis

The data contains 8.88% missingness, meaning all results are based on a reduced sample. The final model reports 8352 (64%) observations were used to estimate the model. This means the results of the logistic regression should be understood with caution, and with knowledge that the model's performance could have improved with complete case data. Meaningful conclusions about the performance and analysis can still be drawn. Using a confusion table revealed the percentage of true positives to be 0.6, and true negatives to be 0.59. As these results are greater than 0.5, the model performs better than chance, although not by much, despite the threshold reflecting the prevalence of problem gamblers in the data. This could be due to the inclusion of *longill12* in the model despite presenting non-significance impacting the ability of the model to distinguish between cases. Furthermore, APC allows for interpretations of the average probability changes with an increase of one-unit in a predictor. The highest APC value was 0.26 for *drating* (alcoholic intake) – a one unit increase in *drating* is associated with a 26% increase in the average predicted probability of being a problem gambler. This highlights a correlation between problem gambling and alcohol.

Lastly, an important consideration is whether any variables have been misreported. Draucker et al. (2020) note people may misreport cigarette habits due to fears of judgement and social stigma. The same can be said for alcohol intake (Fainzang, 2002). If misreporting has occurred, associations between these behaviours and problem gambling could have been different. Despite the discussed limitations of the model and data, none of the results seem counterintuitive or are in disagreement with literature on gambling, providing confidence in the overall direction of the findings.

The Lay Report

This report determines that health outcomes are highly significant in predicting whether someone is a problem gambler. In particular, it has highlighted the diagnosis of a long-standing illness, alcohol intake, cigarette usage, and mental wellbeing as being strongly associated with problem gambling. Alongside this, age, sex, and NS-SEC class prove to be highly predictive of problem gambling, as they are necessarily connected to health outcomes. As the means through which betting has expanded, once from in-person, venue-based practices, to widespread availability through the internet, the range of betting options has expanded beyond traditional sports and casino games to include political and economic events, gambling research is vital. This report shows the key takeaways from this research.

Key Takeaways

1. **Substance Abuse is Telling**
It has been shown that cigarette and drinking habits are associated with problem gambling. This suggests that addressing one form of addiction, such as problem gambling, may require a holistic approach that considers and addresses simultaneous behaviours.
2. **Younger Men are Engaging in Problem Gambling at a High Rate**
This report details a high rate of problem gambling in men, and in young people. This intersection of sex and age leads to the conclusion that young men are particularly vulnerable to problem gambling tendencies. With health proving to be predictive, the impact of mental health is concerning as suicide is reported to be the biggest killer of men under 50 (NHS).
3. **The Ideal of Upward Social Mobility**
As alluded to in the introduction, the means of betting has changed with the expansion of the internet. As Shaffer and Korn (2002) state, poverty is associated with financial risk-aversion, potentially influencing those from lower socioeconomic backgrounds to engage in gambling as a means of improving their financial situation.
4. **Gambling has a Health (and Wider Societal) Impact**
This research has treated health conditions as indicative of problem gambling. This could be understood through these health conditions affecting the likelihood of engaging in problem gambling. However, several studies show gambling has an inverse affect on health (e.g. Black et al., 2013; Fong et al., 2005), leading to substance misuse and depression, and often domestic violence, crime, and suicide. Any causal assumptions should be bidirectional.

Implications of the Findings

The Coalition to End Gambling Ads (CEGA) reports 25% of people who gamble are at risk of harm. This statistic is one of many which supports calls for gambling to be banned or further restricted, therefore policy recommendations cannot be made without demanding measures which lead to the reduced exposure of gambling platforms, dispelling myths of a high chance of success. These systematic policy reforms have been subject to much pushback, and so with regard to the key findings of the report, I recommend there should be increased funding of mental health services to expand the capacity of these services to provide support to current and potential problem gamblers, as well as an increased effort to educate the public of the reality of engaging in gambling.

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Appendix

Appendix A: Merging Variables

I merged the levels of three variables. While this was largely based on natural ordering, I looked to the literature to merge variables reporting GHQ-12 scores into a binary variable, following Goldberg et al.'s binary identification of individuals as 'cases' or 'non-cases' (1997). The details of all merged variables are below, and correspond to code in the accompanying .txt file.

ghq12scr (Goldberg, 1997)

if ghq12 $\geq$ 3 $\rightarrow$ 1,
if ghq12 $<$ 3 $\rightarrow$ 0
else $\rightarrow$ NA

HHSIZE

if HHSIZE = 1 $\rightarrow$ 1
if HHSIZE = 2 $\rightarrow$ 2
if HHSIZE = 3 $\rightarrow$ 3
if HHSIZE = 4 $\rightarrow$ 4
if HHSIZE $>$ 4 $\rightarrow$ 5 (to indicate 5+)
else $\rightarrow$ NA

totinc

if totinc = 1, 2, 3, 4, or 5 $\rightarrow$ 1
if totinc = 6, 7, 8, 9, or 10 $\rightarrow$ 2
if totinc = 11, 12, 13, 14, or 15 $\rightarrow$ 3
if totinc = 16, 17, 18, 19, or 20 $\rightarrow$ 4
if totinc = 21, 22, 23, 24, or 25 $\rightarrow$ 5
if totinc = 26 or 27 $\rightarrow$ 6
if totinc = 28 or 29 $\rightarrow$ 7
if totinc = 30 or 30 $\rightarrow$ 8
if totinc = 96 or 97 $\rightarrow$ NA
else $\rightarrow$ NA